# DA592 Preliminary Term Project Proposal

## 1. Project Title

**FinAgent-Pulse: Multi-Agent Quantitative Trading & Sentiment Analysis Framework using Hybrid RAG and Time-Series Forecasting**

## 2. Team Members

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## 3. Motivation

In modern quantitative finance, relying solely on historical price charts often fails during high-volatility events triggered by news macro-shocks or social media panic. Traditional sentiment bots treat market news superficially, failing to contextualize information through foundational investment principles. Recent advances in Large Language Models allow us to move beyond simple sentiment scoring. By building an Agentic Workflow that combines deep learning time-series models with a Hybrid RAG system, this project aims to simulate a professional institutional investment committee. This creates a scalable, educationally rigorous solution to a multi-billion dollar real-world business problem: mitigating financial risk through contextual market intelligence.

## 4. Project Description

The project defines a multi-layered financial prediction and decision-making problem:
1. The Quantitative Challenge: Predicting short-term stock/index movements using historical market data.
2. The Semantic Challenge: Contextualizing real-time financial texts (news headlines and social media discussions) into a structured sentiment index.
3. The Agentic Challenge: Coordinating multiple specialized AI agents to analyze data, cross-reference findings with regulatory/investment guidelines, and formulate an actionable trading strategy.

We will benchmark and fuse numerical forecasting with semantic analysis to determine whether an asset is facing a genuine market shift or a psychological 'bear/bull trap'.

## 5. Methodology

The architecture consists of three core components deployed via an interactive Streamlit dashboard:

### Data Layer

We will leverage open-source benchmarking financial data from Kaggle, aligning numerical price movements with text embeddings:
• Market Data: Sourced via yfinance API (e.g., Apple, Tesla, or major indices) and structured using paradigms from the Kaggle Two Sigma Financial News dataset.
• Text Data: The Kaggle Reddit Daily News for Stock Market Prediction and Stock Market Sentiment Dataset to historical news corpus.

### Analytical Modeling (ML/DL)

• A Bi-directional LSTM (Long Short-Term Memory) network will process sequence-based temporal numerical data to forecast a 7-day price trajectory.
• FinBERT (Fine-tuned BERT for financial text) will be implemented to generate daily financial sentiment vectors (on a -1 to 1 scale).

### Agentic Framework & Hybrid RAG

We will build an autonomous investment committee using an agent orchestration framework (e.g., CrewAI or LangGraph) consisting of three specialized agents:
1. Data Analyst Agent: Processes LSTM outputs and identifies structural price anomalies.
2. Sentiment Critic Agent: Scans semantic data via a Hybrid RAG system (integrating keyword search, semantic vector search via ChromaDB, and a mini-Knowledge Graph mapping company relations) to extract the core drivers behind the news.
3. Risk Management Agent: Cross-references the findings of the first two agents against a knowledge base of classic investment guidelines (e.g., Benjamin Graham's principles) to issue a finalized 'Buy/Sell/Hold' directive.

## 6. Project Deliverables / Expected Results

• An End-to-End Streamlit Web Application: A fully functional dashboard visualizing real-time and predicted price charts alongside a 'Market Fear & Greed Index' powered by FinBERT.
• AI-Generated Executive Investment Reports: Structured markdown reports generated by the autonomous AI agents explaining the comprehensive why behind their trading strategies.
• Evaluation Metrics: Technical report charting the $R^2$ and RMSE metrics for the LSTM model, accuracy metrics for the sentiment engine, and a comparative ablation study of standard RAG vs. Hybrid RAG performance.

## 7. Milestones & Timeline

The timeline spans across the Summer 2026 term, structured into 2-week sprints as detailed below:

| Phase / Key Deliverable | June 2026 | July 2026 | August 2026 | Target Date |
|---|---|---|---|---|
| Phase 1: Data Pipeline (Ingestion & Preprocessing) | | | | June 05, 2026 |
| Phase 2: Core ML/DL (Bi-LSTM & FinBERT Setup) | | | | June 20, 2026 |
| Phase 3: Hybrid RAG Indexing (ChromaDB & Vector) | | | | July 05, 2026 |
| Phase 4: Agentic Flows & Orchestration Framework | | | | July 20, 2026 |
| Phase 5: UI Development & Dashboard Deployment | | | | August 05, 2026 |
| Phase 6: Evaluation, Benchmarking & Final Report | | | | August 20, 2026 |